KU LEUVEN

Natural Language Processing: Machine translation

H02B1A



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Outline

Introduction

Supervised MT

Decoding with Beam Search

Evaluation
- Surface overlap-based metrics
- Embedding-based metrics

SOTA and open challenges
- State-of-the-art
- Low-resource MT
- Gender bias
- Open challenges

Summary

Source (en)

`Unicorns bring joy to everyone they meet`



Target (nl)

`Eenhoorns brengen vreugde aan iedereen die ze ontmoeten`

Machine translation

The task of translating text from a source language to a target language.

Why is it useful?

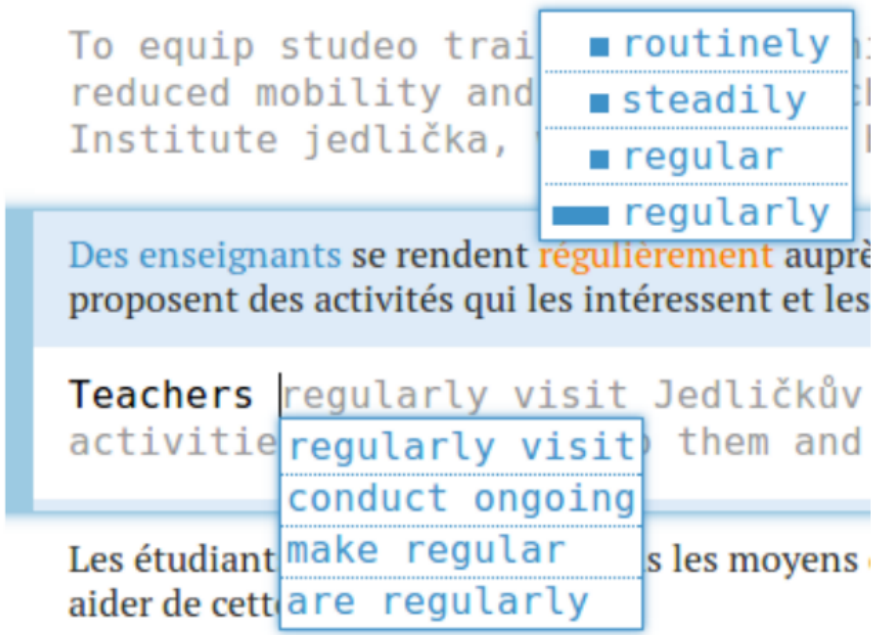


🔗

Information access

- A lot of information on the web is in English
- MT helps making it accessible to non-English speakers.

Why is it useful?



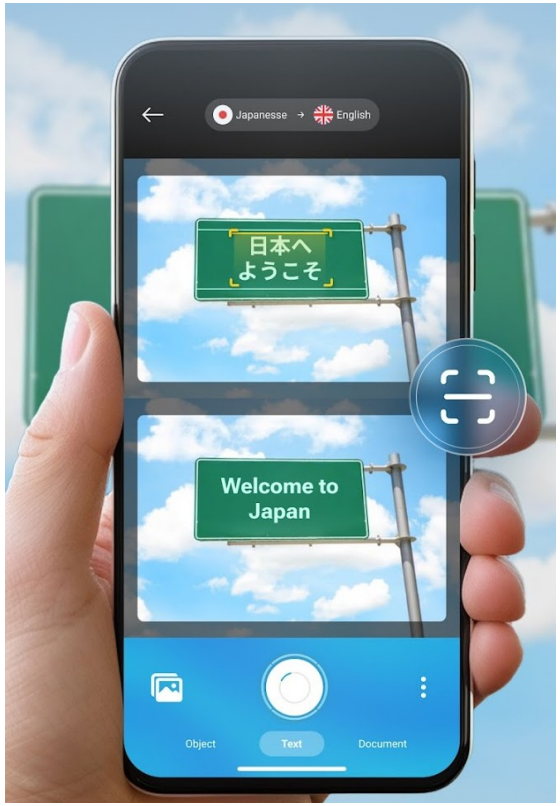
Computer-assisted translation (CAT) can speed up human translation.

Why is it useful?



Speech to speech translation can facilitate direct interaction.

Why is it useful?



Translating images on the fly can facilitate e.g. travelling.

What is difficult?

Word order

Languages differ in how they order words:
E.g. the order of Subject (S) Verb (V) Object (O).

English:	He wrote	a letter	to a friend	SVO
Japanese:	tomodachi *Friend*	ni tegami-o *to letter*	kaita *wrote*	(S)OV
Arabic:	katabt *wrote*	risāla *letter*	li ṡadq *to friend*	VSO

What is difficult?

Words in the source language do not always translate to the same word in the target language.

En →	Nl
Bank	bank
	oever
(verb)	indammen
(verb)	ophopen

(See word sense disambiguation in lecture 9)

What is difficult?

Words in the source language do not always translate to the same word in the target language.

En →	Fr
Dog	chien
	chienne

(Languages vary in morphology, see lecture 7)

What is difficult?

Languages may have lexical gaps wrt other languages.

De →	En
Schadenfreude	The joy a person feels from other people's misfortune.
Wanderlust	The need to travel.
etc.	

What is difficult?

Some languages omit certain words. E.g. pro-drop languages omit certain pronouns.

Spanish: Sofia llegó tarde porque ∅ perdió el autobús.

English: Sofia arrived late because she missed the bus.

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Machine translation requires training data: a parallel corpus.

English

Santa Claus teaches NLP to elves in the North Pole. They learn about machine translation and sentiment analysis. Santa loves training AI models for better toy recommendations. The elves code while enjoying hot cocoa and holiday cheer. Everyone prepares for a data-driven Christmas Eve delivery.

Dutch

De kerstman leert NLP aan elfen op de Noordpool. Ze leren over machinevertaling en sentimentanalyse. De Kerstman traint graag AI-modellen voor betere speelgoedaanbevelingen. De elfen programmeren terwijl ze genieten van warme chocolademelk en kerstsfeer. Iedereen bereidt zich voor op een datagestuurde kerstavondbezorging.

First step: sentence alignment.

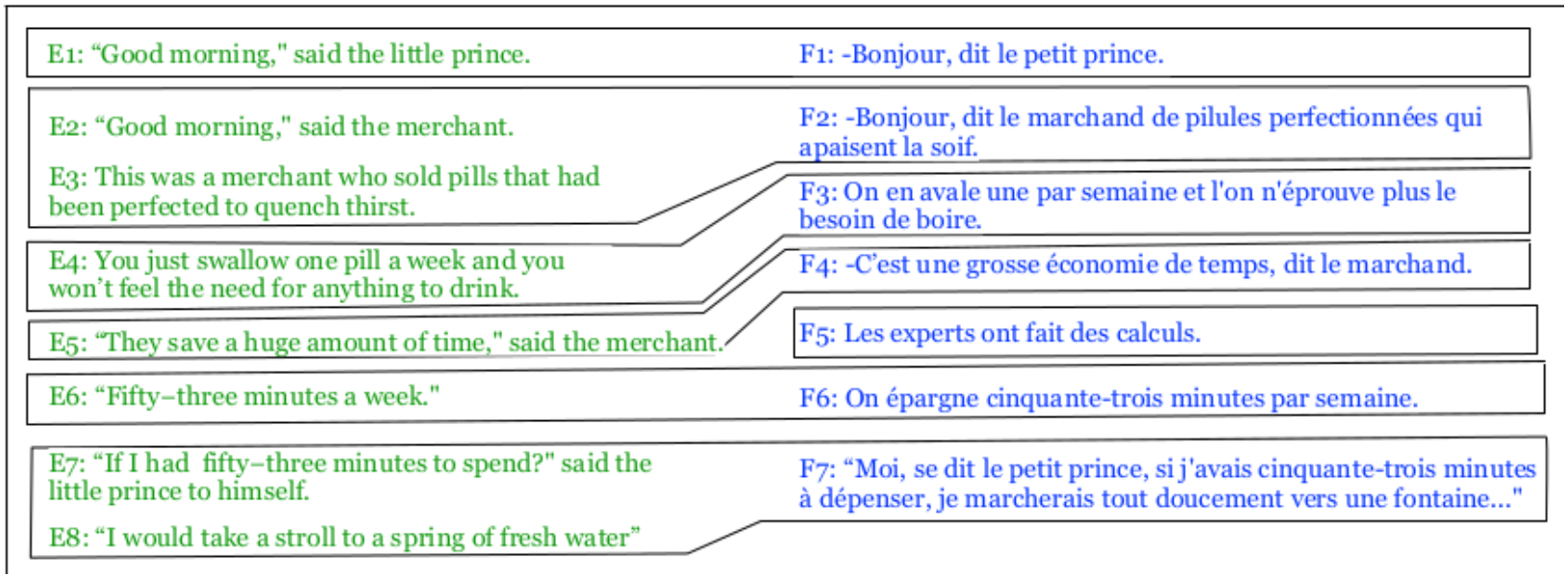
English

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2. They learn about machine translation and sentiment analysis.
3. Santa loves training AI models for better toy recommendations.
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5. Iedereen bereidt zich voor op een datagestuurde kerstavondbezorging.

It is not always that easy.



Sample alignment of *Le Petit Prince* in English and French (Jurafsky and Martin, 2025)

Not always a one-to-one mapping between sentences.

Sentence alignment

1. Score pairs of spans to measure how likely they are to be translations
   - Using e.g. character lengths
   - Using a multilingual embedding space and cosine similarity
2. Align groups of up to 2 sentences based on these scores, using dynamic programming

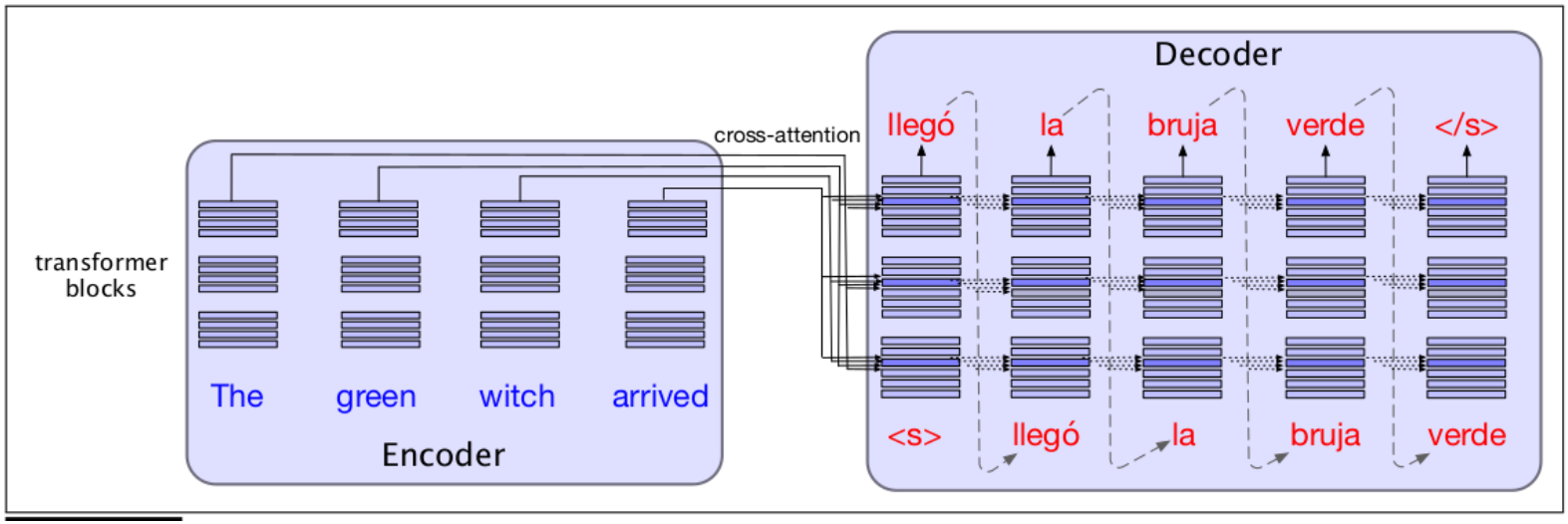
MT as a supervised task

- Given a parallel corpus of sentences tokenized (into subwords)
- Maximize the probability of the sequence of tokens in the target language $y_1, ..., y_m$, given the sequence of tokens in the source language $x_1, ..., x_n$:

$$p(y_1, ..., y_m | x_1, ..., x_n)$$

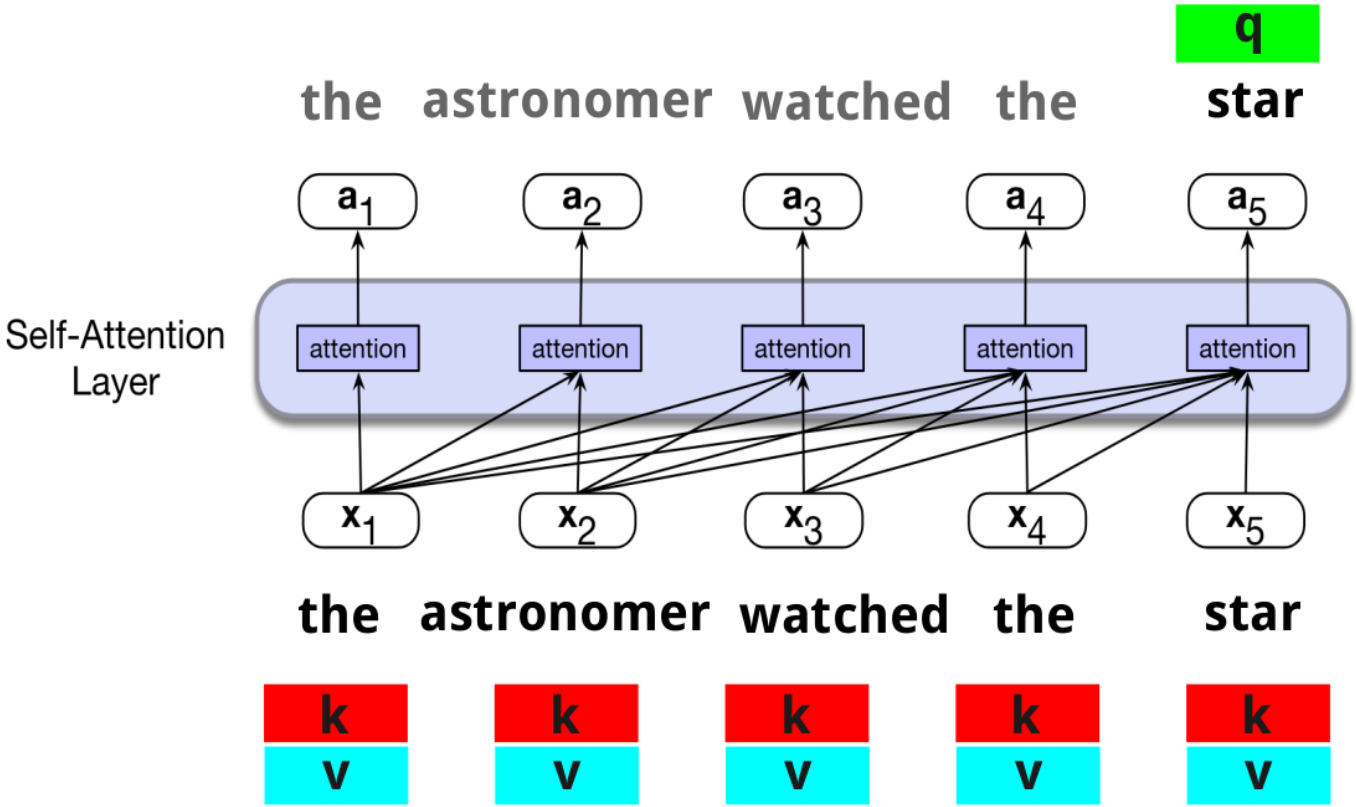
- Modelled via an encoder-decoder architecture:

$$\boldsymbol{h} = encoder(x)$$
$$y_{t+1} = decoder(\boldsymbol{h}, y_1, ..., y_t) \forall t \in [1, ..., m])$$

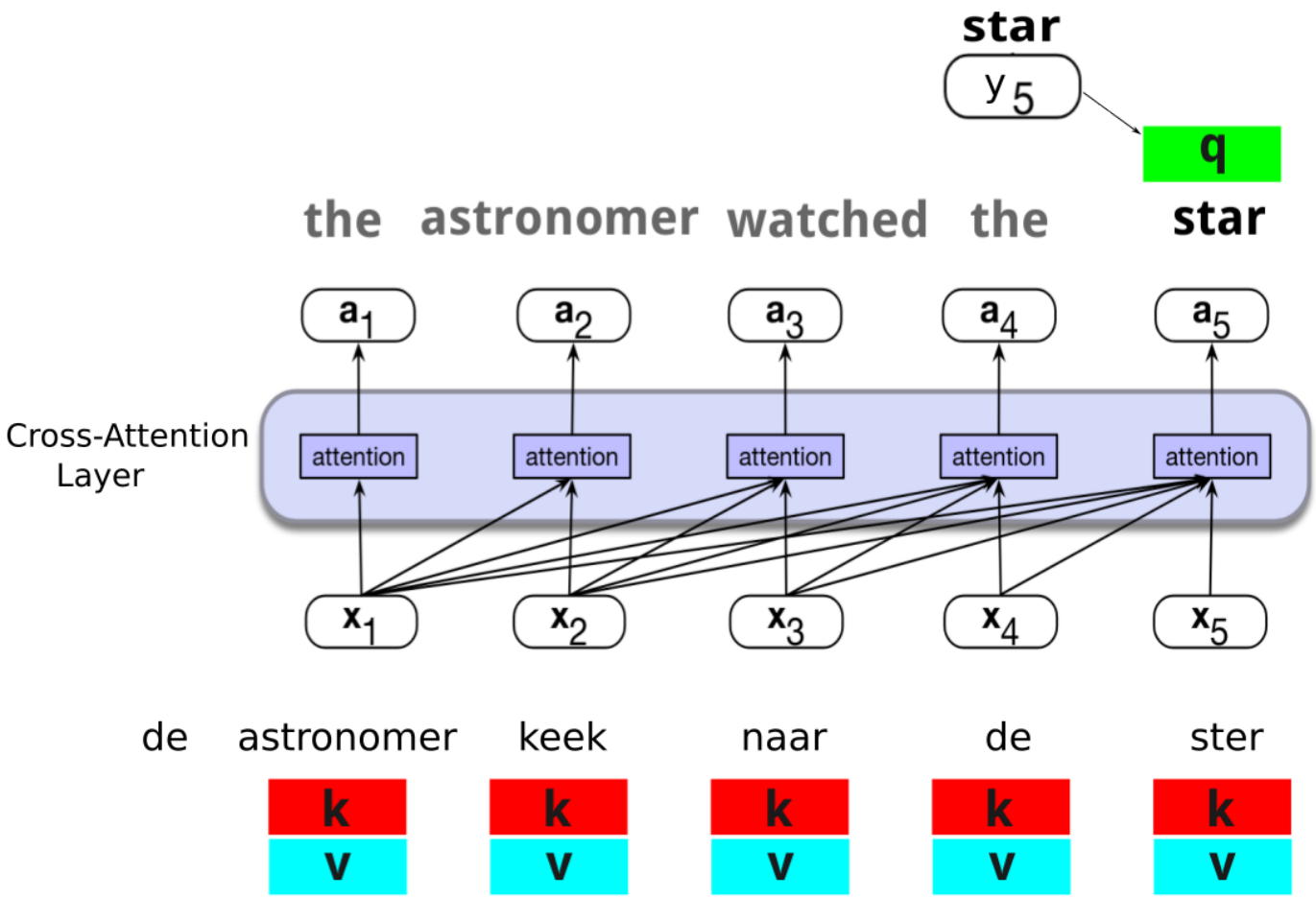


The encoder-decoder transformer architecture for machine translation (Jurafsky and Martin, 2025).

Remember self-attention? (Lecture 4)

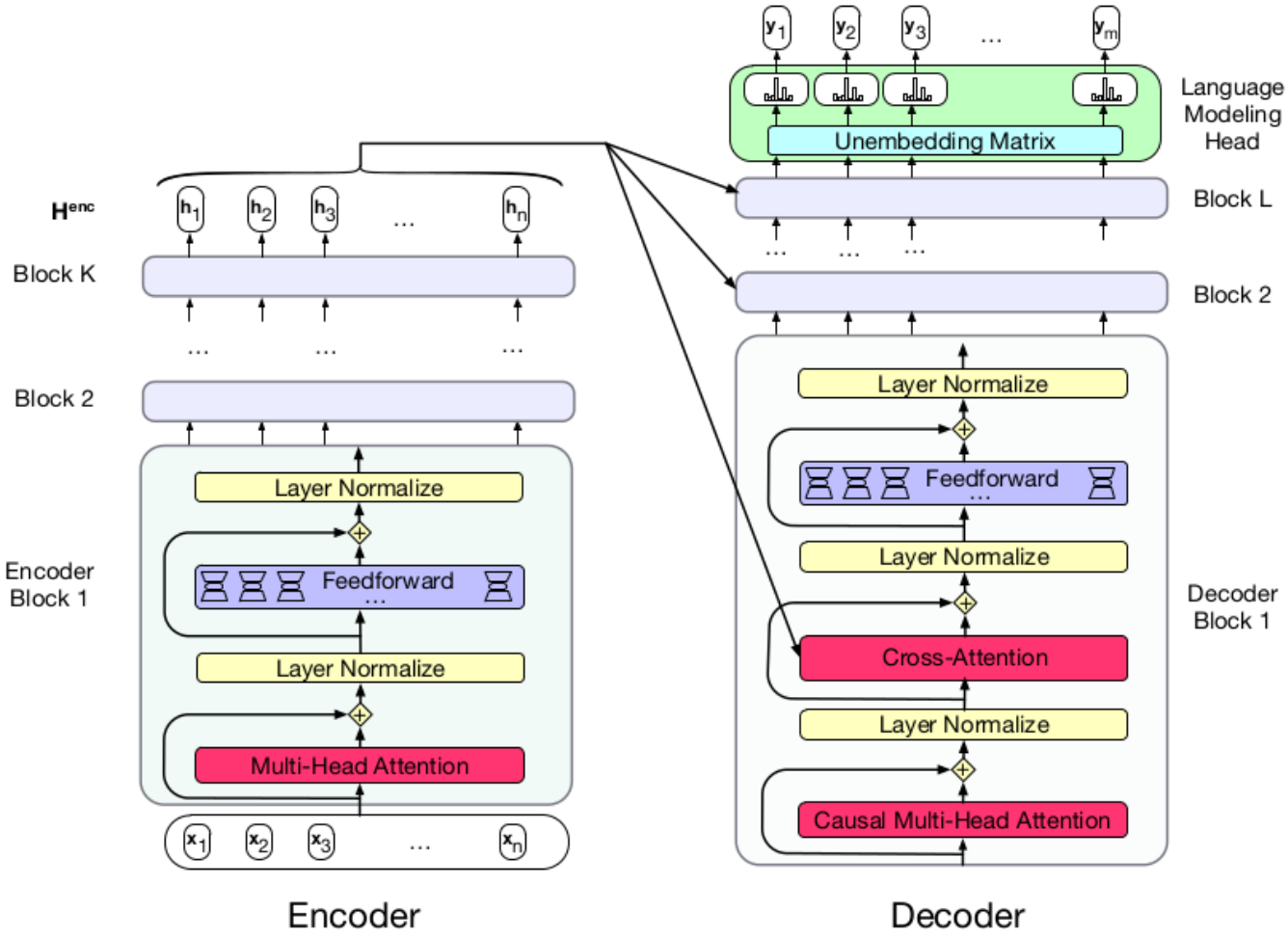


Self-attention (Jurafsky and Martin, 2025).

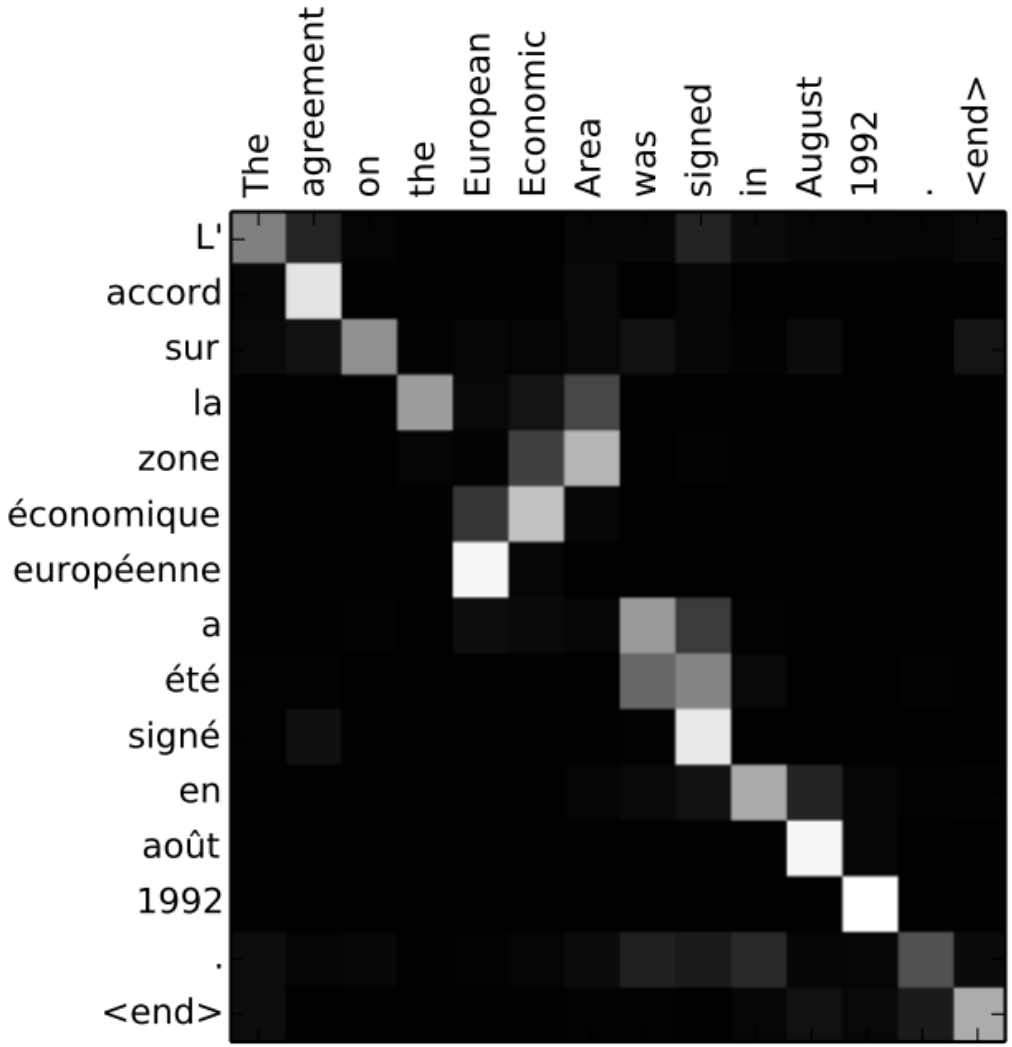


Cross-attention, adapted from Jurafsky and Martin (2025).

Cross-attention uses the same mechanism but the keys and values come from the output of the encoder.



Encoder-decoder transformer blocks (Jurafsky and Martin, 2025).



Visualizing attention weights when translating a French sentence into English. 🔗

- Just like for language modelling, the model is trained to predict the next word.
- Cross-entropy loss
- Using teacher forcing

See lecture 4

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Greedy decoding

- At each time step t, choose the highest probability word:

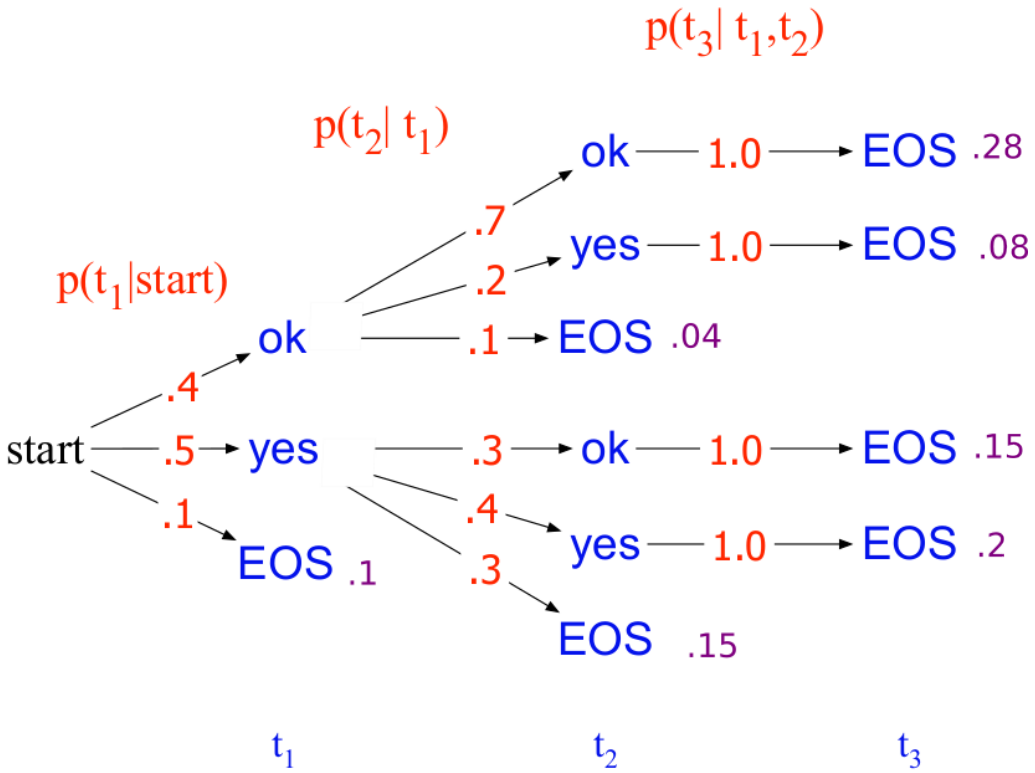
$$\hat{w}_t = \arg\max_{w \in V} P(w | \boldsymbol{w}_{1:t})$$

- May lead to globally suboptimal choices

$\rightarrow$ Beam search maintains multiple choices and leaves choice to later

Beam search

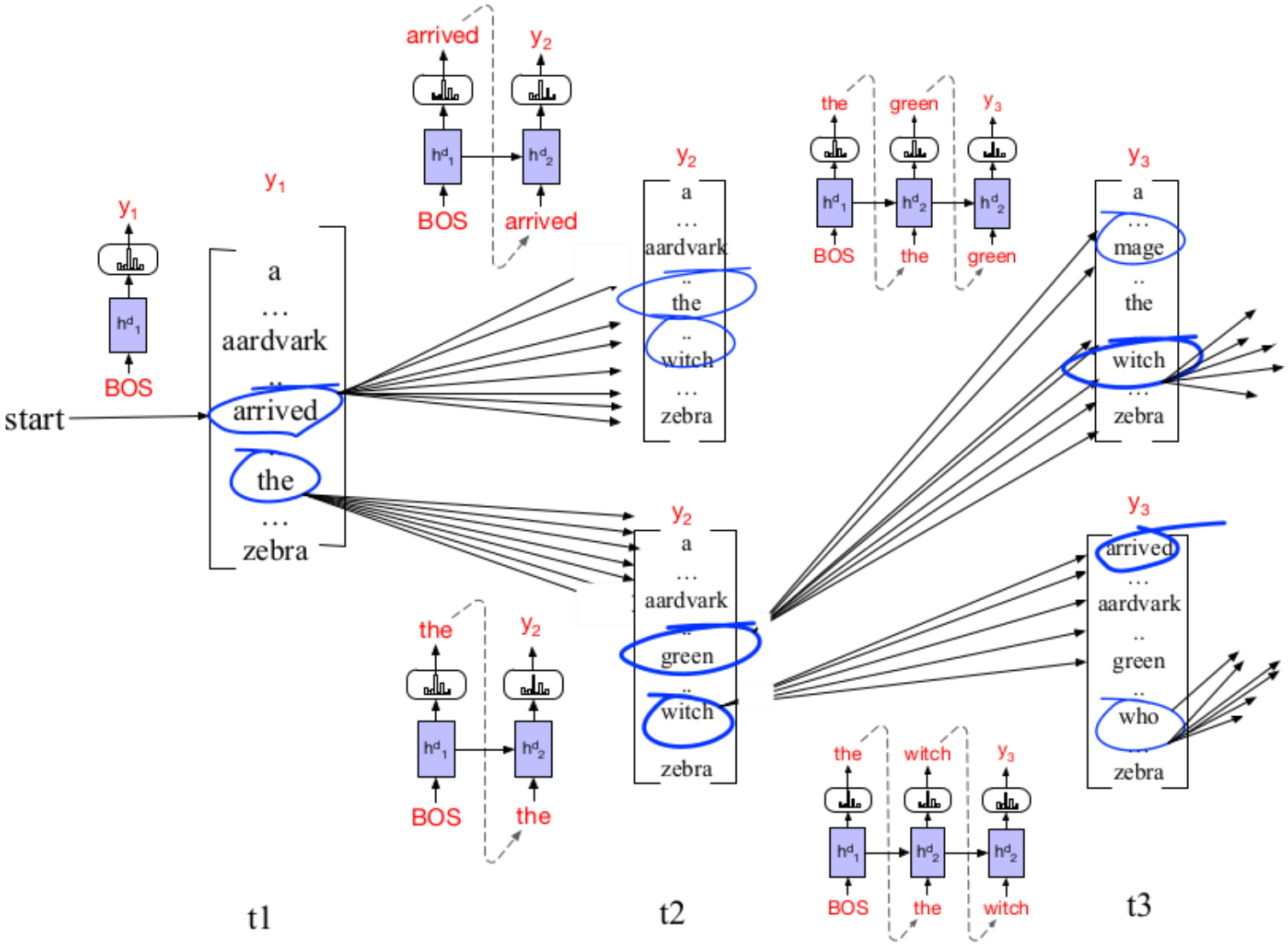
Search for the best path in a tree.



Search tree for language modelling (Jurafsky and Martin, 2025)

Beam search

- Search for the best path in a tree.
- Exhaustive search is too slow.
- Instead, keep k possible tokens at each step
- k is the beam width



Beam search decoding with $k = 2$ (Jurafsky and Martin, 2025).

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Evaluation

Evaluation dimensions

Adequacy: How well the translation captures the exact meaning of the source sentence.

Fluency: How fluent the translation is in the target language (clarity, grammaticality)

Human evaluation

- Rating
  - Fluency
  - Adequacy:
    - How much information is preserved from source to target
    - Comparing to a reference translation
    - Ranking candidate translations
- Post-edit the translation and measure the difference

$\rightarrow$ Best evaluation but time consuming and expensive

Automatic evaluation

Comparison between a human reference translation and a candidate MT translation

- chrF: character overlap
- BLEU: word overlap
- Embedding-based methods

chrF

- Given a hypothesis candidate MT translation and a reference translation,
- the character F-score measures the character overlap,
- parametrized by k: the max length of the n-grams to consider.

chrF

chrP: percentage of character 1-grams, 2-grams ... k-grams in the hypothesis that occur in the reference, averaged.

chrR: percentage of character 1-grams, 2-grams ... k-grams in the reference that occur in the hypothesis, averaged.

chrF+β:

$$(1+\beta^2)\frac{chrP \times chrR}{\beta^2 \times chrP + chrR}$$

chrF2: (common)

$$\frac{5 \times chrP \times chrR}{4 \times chrP + chrR}$$

chrF example $k = \beta = 2$

REF `witnessforthepast`, unigrams: 18, bigrams: 17

HYP `witnessofthepast`, unigrams: 17, bigrams: 16

✔ 1-grams: 17

✔ 2-grams: 13

chrP: $\frac{1+.813}{2} = .906$

P 1-gram: $\frac{17}{17} = 1$

P 2-gram: $\frac{13}{16} = .813$

note: the comma is counted as a character

chrF example $k = \beta = 2$

REF `witnessforthepast`, unigrams: 18, bigrams: 17

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✔ 1-grams: 17

✔ 2-grams: 13

chrP: $\frac{1+.813}{2} = .906$

chrR: $\frac{.944+.765}{2} = .855$

R 1-gram: $\frac{17}{18} = .944$

R 2-gram: $\frac{13}{17} = .765$

chrF example $k = \beta = 2$

REF `witnessforthepast`, unigrams: 18, bigrams: 17

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✔ 1-grams: 17

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chrP: $\frac{1+.813}{2} = .906$

chrR: $\frac{.944+.765}{2} = .855$

chrF: $5\frac{chR \times chR}{4chrP+chrR} = .86$

BLEU

- Based on overlap of word *n*-grams
- Using precision only:
  - Percentage of word 1-grams, 2-grams, 3-grams and 4-grams in the hypothesis that occur in the reference.
  - This is calculated across all the sentences of the corpus.
- Using a brevity penalty:
  - penalizing translations that are too short
- Note: sensitive to word tokenization!

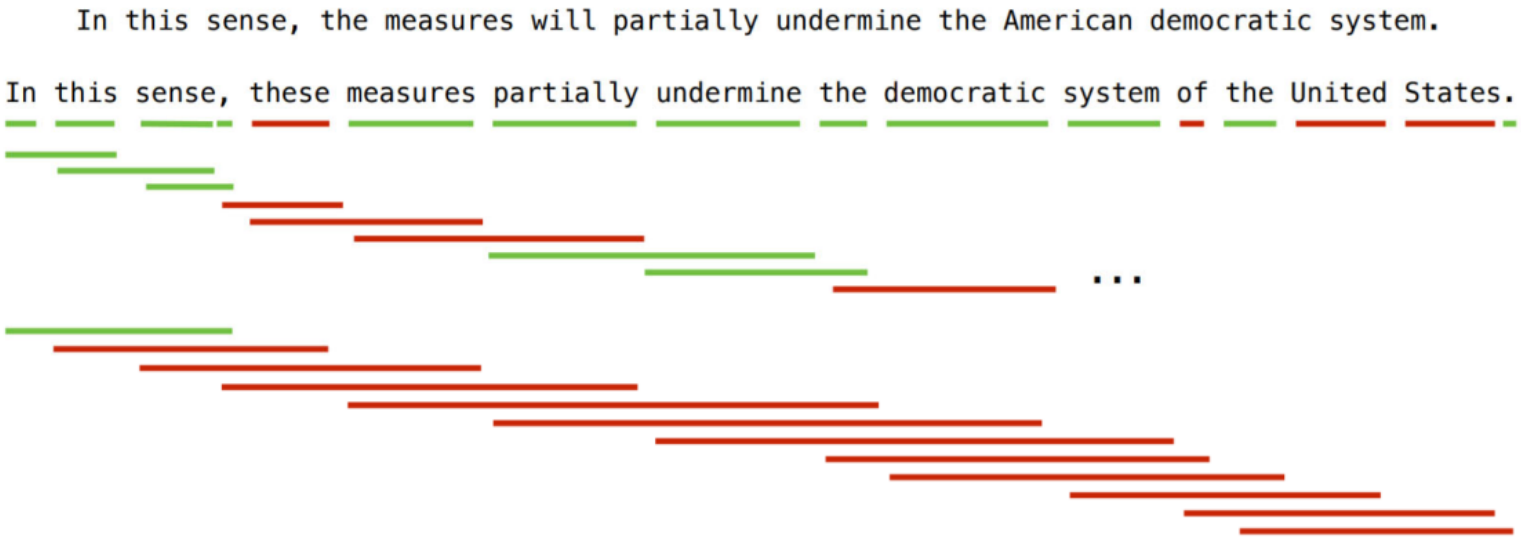


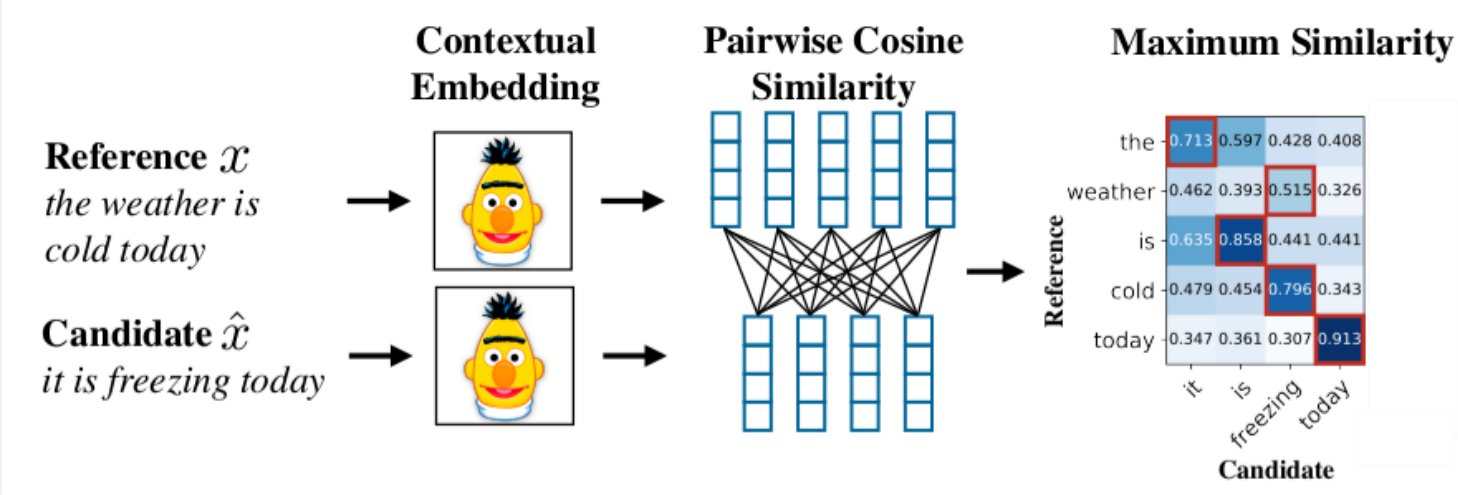
Illustration of $n-$gram overlap accuracies as part of computing BLEU. Image by Dan Klein 🔗.

Limitations of chrF and BLEU

- Synonymy (or more generally, word meaning) is not captured (see lecture 9)
- Word order differences are not captured
- Cross-sentence properties are not captured (e.g. discourse coherence, see lecture 10).
- Generally, highly criticized but still heavily used for lack of (cheap) alternatives

Embedding-based evaluation

- Given a hypothesis translation, a reference translation and a rating
  - Fine-tune an encoder to predict the rating
- Given a hypothesis translation, a reference translation:
  - Compute embeddings of the hypothesis and the reference using an encoder
  - Measure the similarity of the embeddings



BertScore visualization (Jurafsky and Martin, 2025).

R : average max similarity per reference word

P : average max similarity per candidate word

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SOTA

Quality tiers

Publishable: Publish as is HTER: 0-10%

Editable: Requires post-editing HTER: 10-20%

Gistable: Useful for information gathering or trend spotting HTER: 20-40%

Triagable: Only useful for identifying relevant documents HTER: 40-50%

Tiers defined based on human-annotator translation editor rate (HTER).

The scale was developed in preparation of the DARPA GALE programme.

SOTA

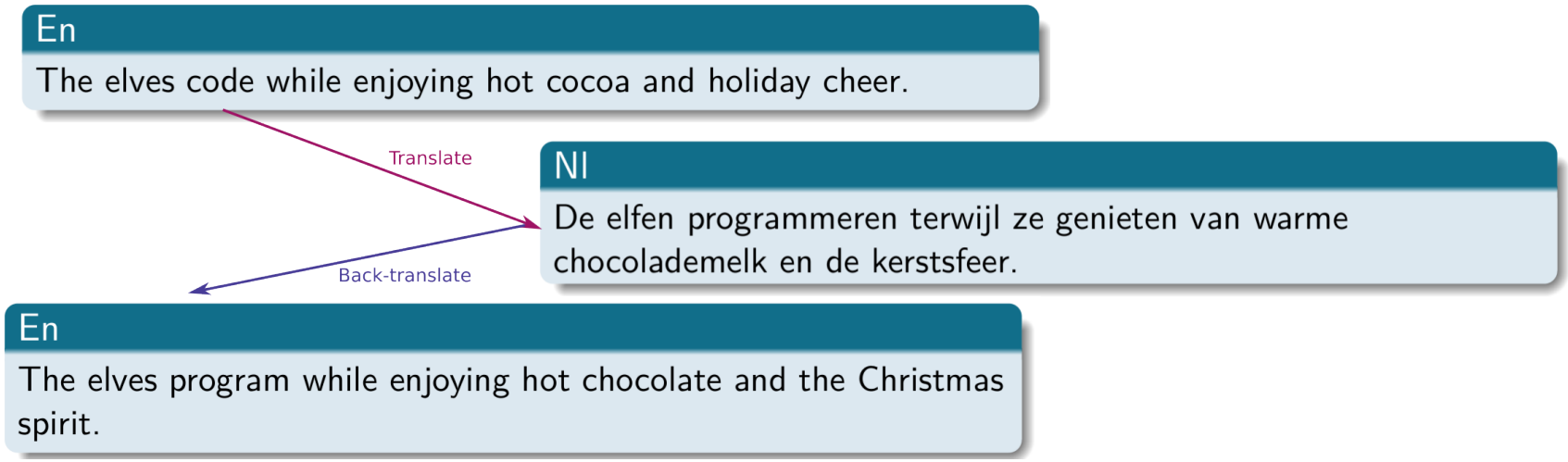
Language pairs and domains	assessment
French-English restricted domain	publishable
French-English news stories	publishable
German-English news stories	publishable
Swahili-English news stories	gistable
Uyghur-English news stories	triagable

Informal estimates by Philipp Koehn. 🔗

- MT relies on large parallel corpora
- Only available for a fraction of possible language pairs
- What can we do in low-resource settings?

Backtranslation

A type of data augmentation to add sentence pairs to the training data.



Multilingual models

One model, many translation directions.

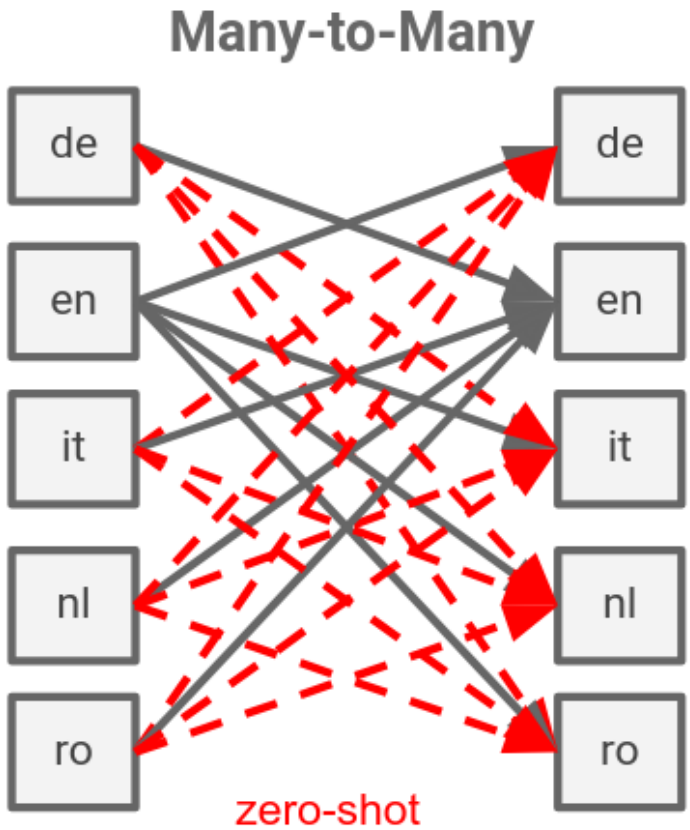


Illustration of a many-to-many machine translation model (Aharoni et al., 2019).

Multilingual models

- One model for 100s of language pairs (Aharoni et al., 2019).
- Enables cross-lingual transfer between related languages (e.g. Spanish and Catalan).
- Empirically performs competitively for some language pairs.

Low-resource MT

Limitations and open challenges

- Low-resource MT is still very much an open challenge
- NMT models are data hungry
- Certain language properties make MT more difficult:
  - Morphological complexity (e.g. Ataman and Federico (2018))
  - Word order freedom (Bisazza et al., 2021)
- Data quality can be an issue (Kreutzer et al., 2022)
- English almost always included in language pairs under study

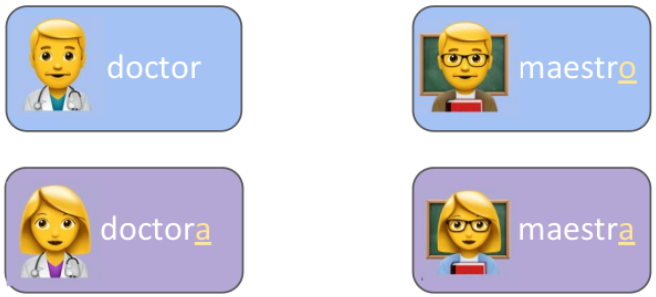
Low-resource MT

Limitations and open challenges

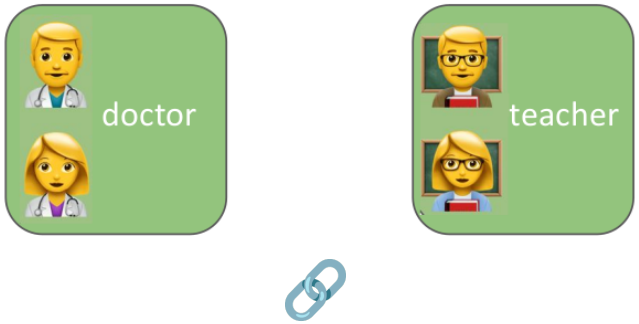
AmericasNLP shared task (De Gibert et al., 2025) involving polysynthetic languages:

- Poor baselines, hard to beat
- LLMs help if target side is high-resource, otherwise not.

Some languages encode grammatical gender (Spanish, Italian, Russian, ...)



Others do not (English, Turkish, Basque, Finnish)



In some cases, this can lead to two possible translations.

En

The doctor asked for help.

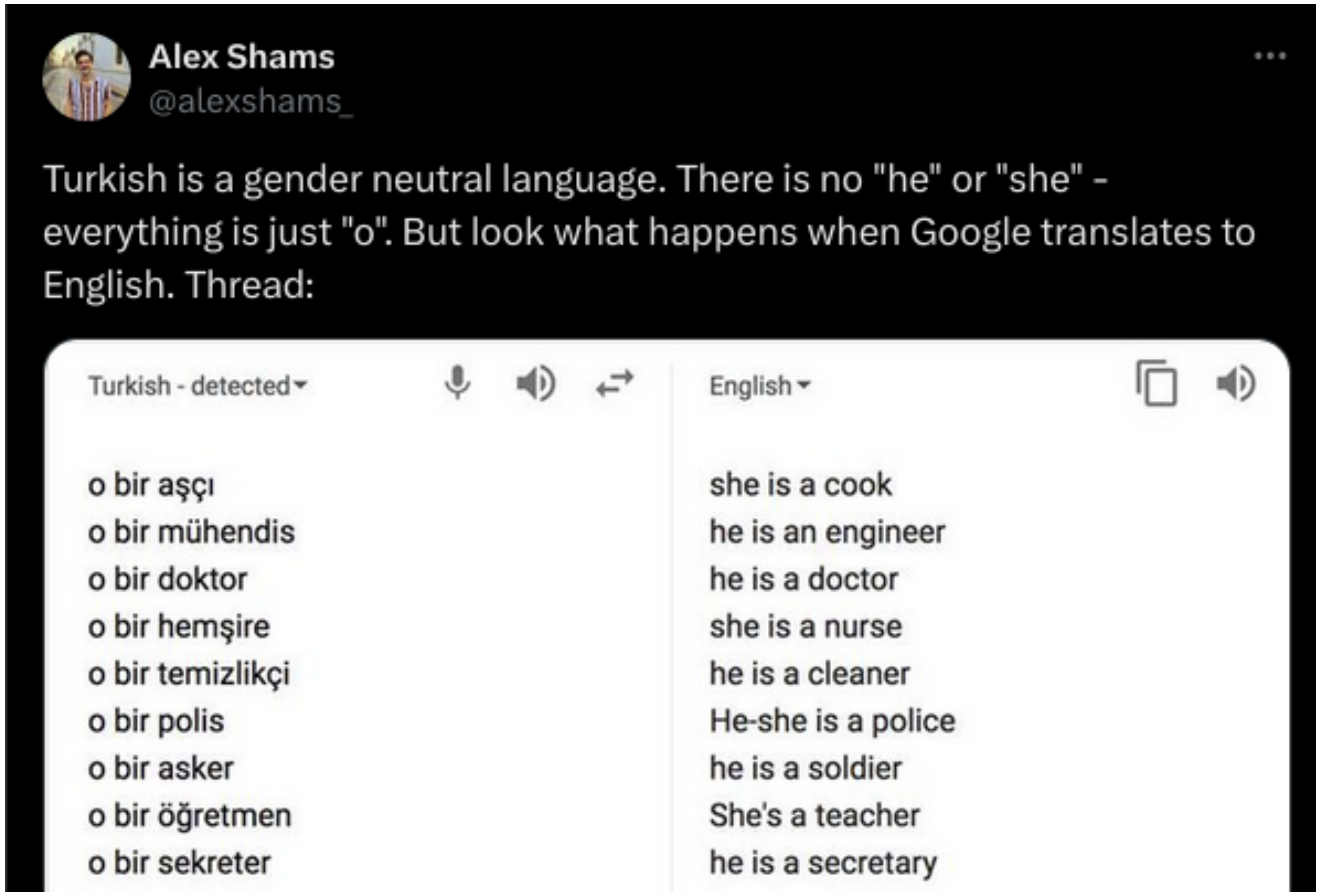
↓

Es

👩‍⚕️ La doctora pidió ayuda.

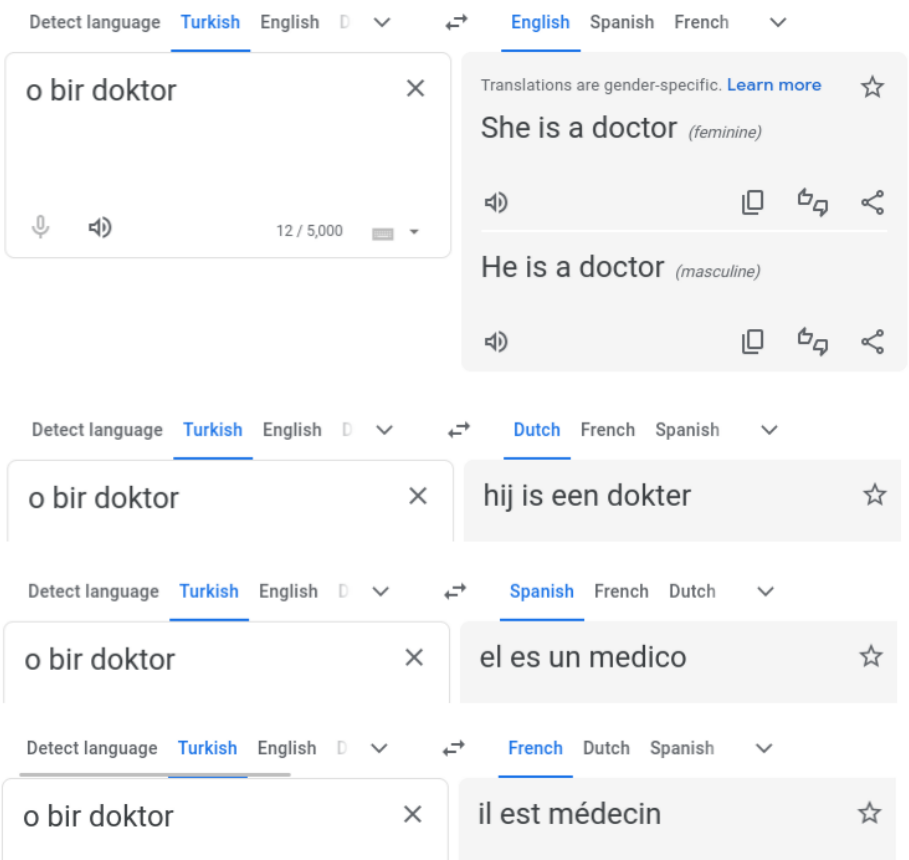
👨‍⚕️ El doctor pidió ayuda.

Machine translation tools often stick to one translation.



Xeet from 2017 🔗

🙋 But it's fixed now - right? 🤦‍♀️ Well...



Screenshot from 27 Nov 2024

In other cases, there is only one possible translation.

En

The doctor asked the nurse to help her in the procedure.

↓

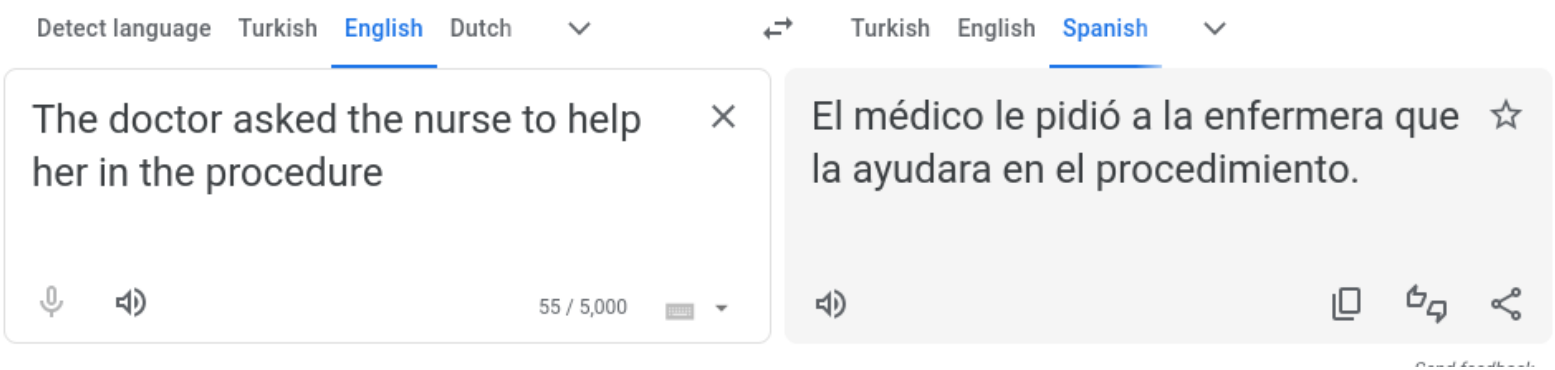
Es



La doctora le pidió a la enfermera que le ayudara con el procedimiento.

Gender bias may lead to errors.

Gender bias may lead to errors.



Screenshot from 27/11/2024

Gender bias evaluation

- Constructing stereotypical vs non-stereotypical examples
- Testing MT on both types of examples
- Measuring gender translation accuracy

Non-stereotypical

En

The doctor asked the nurse to help her in the procedure.

↓

Es



La doctora le pidió a la enfermera que le ayudara con el procedimiento.

Stereotypical

En

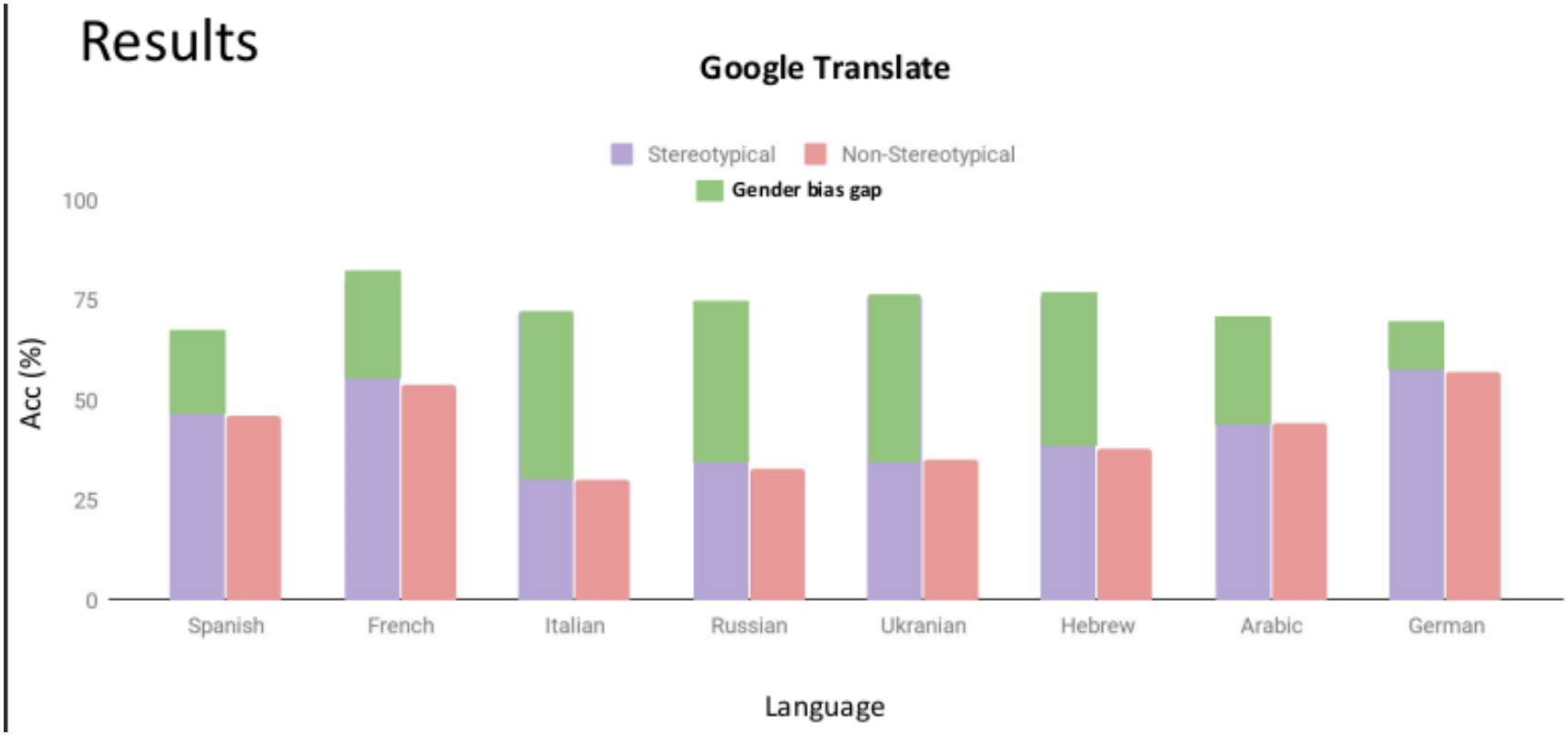
The doctor asked the nurse to help him in the procedure.

↓

Es



El doctor le pidió a la enfermera que lo ayudara con el procedimiento.



Results on gender translation accuracy. 🔗

- Gender bias in MT is still an open challenge.
- There are some methods to mitigate it, e.g. using controlled text generation (Lu et al., 2023).
- There is a workshop dedicated to this, *Gender Inclusive Translation Technologies* GITT.

Open challenges

- Low-resource MT
- Gender bias
- Document-level translation (Läubli et al., 2018)
- Efficient improvements over beam search
  - Minimum Bayes Risk (MBR) is computationally expensive (Trabelsi et al., 2024)
- Low-frequency words
- Out-of-domain data

See also Koehn and Knowles (2017), still largely relevant today.

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- Machine translation with a neural encoder-decoder model
- Decoding with beam search
- Automatic evaluation methods and their limitations
- State-of-the art: highly dependent on language pairs and domain
- Strategies for low-resource MT
- Gender bias in MT
- Overall, much progress but big open challenges

Acknowledgments

The quality tiers slides were inspired by the MT course by Philipp Koehn. Good resource if you want to dig more into the topic!
The gender bias slides were inspired by Gabriel Stanovsky: 🔗.

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